

PAPER416; $T_s^{\mu\nu}$ Full Five-Component Stress-Energy Decomposition: SCm, UA, Solar Wind, Stellar, and Luminosity Terms

Session: 110 (grokshare755feea7.txt analysis)

PAPER416 – $T_s^{\mu\nu}$ Full Five-Component Stress-Energy Decomposition: SCm, UA, Solar Wind, Stellar, and Luminosity Terms

Source: grokshare755feea7.txt — "Star Magic" Chapter 7 + computeAmunu sections **Session:** 110 (grokshare_755feea7.txt analysis) **CP4** **Class:** TsUniverse5ComponentStressEnergyDecompositionCalculator (#66)

1. Overview

PAPER416 expands the basic two-component stress-energy tensor $T_s^{\mu\nu}$ (covered in PAPER406, which treats only stellar density + luminosity) to the **full five-component decomposition** that includes solar wind, SCm, and UA contributions. This is the complete $T_s^{\mu\nu}$ entering the UQFF metric tensor $A_{\mu\nu}$.

2. PAPER406 vs PAPER416

Component	PAPER_406 (basic)	PAPER_416 (full)
Stellar rest energy	$\blacksquare M_s c^2 / V$	$\blacksquare$
Luminosity correction	$\blacksquare L_s / (c^2 V)$	$\blacksquare$
Solar wind kinetic	$\blacksquare$	$\blacksquare \rho_{sw} v_{sw}^2$
SCm kinetic	$\blacksquare$	$\blacksquare \rho_{\text{SCm}} v_{\text{SCm}}^2 / c^2$
UA kinetic	$\blacksquare$	$\blacksquare \rho_A v_{\text{UA}}^2 / c^2$

3. Full Five-Component $T_s^{\mu\nu}$

$$\boxed{T_s^{\mu\nu} = \frac{M_s c^2}{V} + \frac{L_s}{c^2 V} + \rho_{sw} v_{sw}^2 + \frac{\rho_{\text{SCm}} v_{\text{SCm}}^2}{c^2} + \frac{\rho_A v_{\text{UA}}^2}{c^2}}$$

where $V \approx \frac{4}{3}\pi R_s^3$ is the stellar volume.

3.1 Individual Term Evaluation (Sun)

Term 1 — Stellar rest energy density:

$$\frac{M_s c^2}{V} = \frac{1.989 \times 10^{30} \times 9 \times 10^{16}}{\frac{4}{3}\pi (6.96 \times 10^8)^3} = \frac{1.79 \times 10^{47}}{1.41 \times 10^{27}} \approx 1.27 \times 10^{20} \text{ J/m}^3$$

Term 2 — Luminosity correction:

$$\frac{L_{\odot}}{c^2 V} = \frac{3.828 \times 10^{26}}{9 \times 10^{16} \times 1.41 \times 10^{27}} \approx \frac{3.83 \times 10^{26}}{1.27 \times 10^{44}} \approx 3.01 \times 10^{-18} \text{ J/m}^3$$

Term 3 — Solar wind kinetic density:

$$\rho_{\text{sw}} v_{\text{sw}}^2 = 8 \times 10^{-21} \times (5 \times 10^5)^2 = 8 \times 10^{-21} \times 2.5 \times 10^{11} = 2 \times 10^{-9} \text{ Pa}$$

Term 4 — SCm kinetic energy density:

$$\frac{\rho_{\text{SCm}} v_{\text{SCm}}^2}{c^2} = \frac{10^{15} \times 10^{16}}{9 \times 10^{16}} \approx 1.11 \times 10^{14} \text{ J/m}^3$$

Term 5 — UA kinetic energy density:

$$\frac{\rho_A v_{\text{UA}}^2}{c^2} = \frac{10^{-23} \times (10^8)^2}{9 \times 10^{16}} \approx 1.11 \times 10^{-24} \text{ J/m}^3$$

3.2 Dominant Terms

Ranking by magnitude:

SCm term: $\sim 1.11 \times 10^{14} \text{ J/m}^3 \leftarrow \text{dominant}$

Stellar rest energy: $\sim 1.27 \times 10^{20} \text{ J/m}^3 \leftarrow \text{even larger}$ (but depends on V)

All others negligible by many orders of magnitude

*The text notation $T_{s^{00}} \approx 1.27 \times 10^3 + 1.11 \times 10^7$ from the book refers to **normalized units** where masses/densities are expressed per unit $M_{\odot} \cdot c^2 / V_{\odot}$ — the relative scaling of these two terms determines the physics.*

4. Metric Correction Tensor $A_{\mu\nu}$

The UQFF metric tensor incorporating $T_{s^{\mu\nu}}$:

$$A_{\mu\nu} = g_{\mu\nu} + \eta \cdot T_{s^{\mu\nu}}$$

In the static diagonal approximation:

$$A_{\mu\nu} \approx \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} + \eta \cdot \begin{pmatrix} T_{s^{00}} & & & \\ & -T_{s^{11}} & & \\ & & -T_{s^{22}} & \\ & & & -T_{s^{33}} \end{pmatrix}$$

For the Sun with $\eta = 10^{-22}$:

$$\eta \cdot T_{s^{00}} (\text{stellar rest}) \approx 10^{-22} \times 1.27 \times 10^{20} \approx 1.27 \times 10^{-2}$$

$$\eta \cdot T_{s^{00}} (\text{SCm}) \approx 10^{-22} \times 1.11 \times 10^{14} \approx 1.11 \times 10^{-8}$$

Full numerical metric perturbation:

$$A_{00} \approx 1 + 1.27 \times 10^{-2} + 1.11 \times 10^{-8} + \mathcal{O}(10^{-20})$$

5. Relevance to FU

$A_{\mu\nu}$ enters F_U as the spacetime curvature term:

$$F_U \ni (g_{\mu\nu} + \eta \cdot T_s^{\mu\nu}([\text{UA}], [\text{SCm}], \rho_A, t_n))$$

The full energy budget of a star encoded in $T_s^{\mu\nu}$ — from SCm superconductor to solar wind — is thereby present in every UQFF computation.

6. Code: Stress-Energy in CelestialBody

```
// Stress-energy terms in main.cpp / CelestialBody.cpp
struct TensorT {
double T00_stellar; // Ms * c^2 / Volume
double T00_luminosity; // Ls / (c^2 * Volume)
double T00_solarwind; // rho_sw * v_sw^2
double T00_SCm; // rho_SCm * v_SCm^2 / c^2
double T00_UA; // rho_A * v_UA^2 / c^2
double total() const {
return T00_stellar + T00_luminosity + T00_solarwind + T00_SCm + T00_UA;
}
};

TensorT compute_T_s00(const CelestialBody& body, double rho_A, double v_UA,
double rho_SCm, double v_SCm, double rho_sw, double v_sw) {
const double c = 3e8;
double V = (4.0/3.0) * M_PI * pow(body.Rs, 3);
TensorT T;
T.T00_stellar = body.Ms * c * c / V;
T.T00_luminosity = body.Ls / (c * c * V);
T.T00_solarwind = rho_sw * v_sw * v_sw;
T.T00_SCm = rho_SCm * v_SCm * v_SCm / (c * c);
T.T00_UA = rho_A * v_UA * v_UA / (c * c);
return T;
}
```

7. Unit Tests

```
def test_T00_SCm_dominant():
    """SCm term exceeds UA term by 38 orders of magnitude"""
    c = 3e8
    T_SCm = 1e15 * (1e8)**2 / c**2
    T_UA = 1e-23 * (1e8)**2 / c**2
    ratio = T_SCm / T_UA
    assert ratio > 1e37

def test_A_mu_nu_correction():
    """Metric correction is small (eta = 1e-22, T_SCm ~ 1.11e14)"""
    eta = 1e-22; T_SCm = 1.11e14
    delta = eta * T_SCm
    assert delta < 0.1, f"Metric perturbation must be <<1, got {delta}"
```